

Designing an Agentic AI Alpha Researcher

Alec Hewitt

Prototype research system

Presentation Overview

Task (Topic B) Overview

Input Data

Architecture Overview

Researcher Agent

Evaluation Engine

Critic Agent

Improver Agent

Example

Possible Improvements

Conclusion

Task (Topic B)

Objective

Build an agentic research system that can **propose, evaluate, critique, and revise** quantitative alpha hypotheses using empirical feedback.

Research Agent: Propose → **Code: Evaluate** → **Critic Agent: Critique**
→ **Improver Agent: Revise**

Inputs

- ▶ Engineered market features (volatility_20d, relative_volume, etc.)
- ▶ Proposed signals evaluated against realized 5-day forward returns
- ▶ OpenAI API for agent reasoning (5.6-Luna)

Outputs

- ▶ Economic thesis
- ▶ Executable alpha signal definition
- ▶ Quantitative evaluation metrics
- ▶ Agent iteration history
- ▶ Final validation / test decision

Data and Feature Construction

Raw daily market data

Field	Meaning
date	trading date
ticker	stock identifier
close	daily closing price
volume	daily trading volume

One observation corresponds to a **stock–date pair**.

Prediction target

$$r_{i,t}^{(5)} = \frac{P_{i,t+5}}{P_{i,t}} - 1$$

Future 5-day return is used only as the evaluation target, not as an input to the signal.

Engineered features available to the agent

$$\text{return}_{5d} = \frac{P_t}{P_{t-5}} - 1$$

Short-term trailing price return

$$\text{momentum}_{20d} = \frac{P_t}{P_{t-20}} - 1$$

Medium-term trailing price momentum

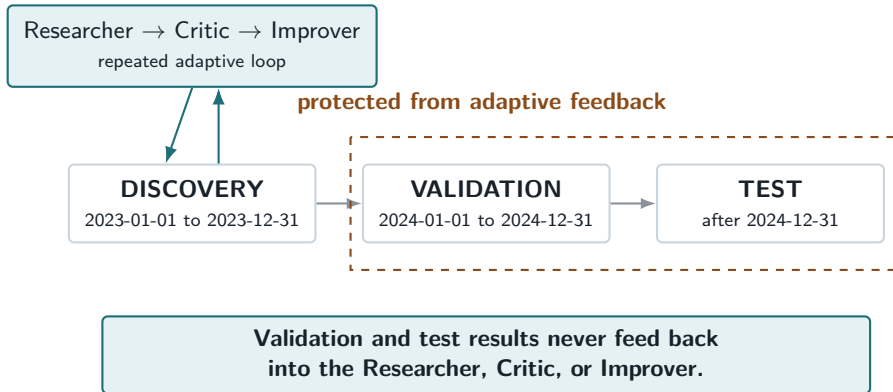
$$\text{volatility}_{20d} = \text{Std}(r_{t-19}, \dots, r_t)$$

Recent realized return volatility

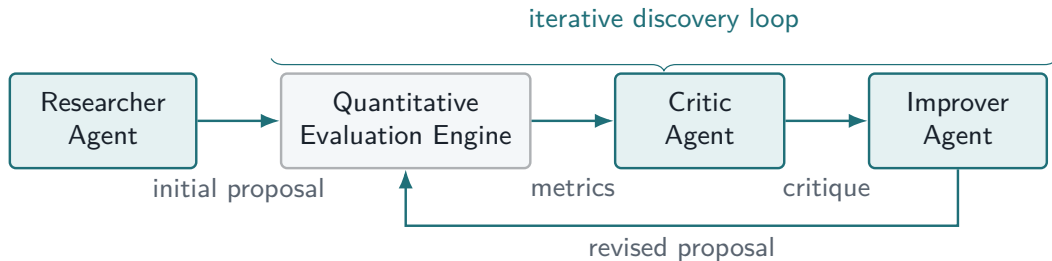
$$\text{relative volume} = \frac{V_t}{\overline{V}_{20d}}$$

Current volume relative to recent average volume

System Architecture



Agentic Loop



Researcher Agent: Generate the Initial Hypothesis

Purpose

The Researcher Agent generates the **initial alpha proposal** that seeds the iterative research loop.

Information provided

► Available engineered features:

- `return_5d`
- `momentum_20d`
- `volatility_20d`
- `relative_volume`

► Allowed operations:

$+$, $-$, $\times$, $\div$

Output: initial proposal

The Researcher must return:

- two features: x_1, x_2
- operation f
- weights w_1, w_2
- economic thesis

$$\alpha_{i,t} = f(w_1 x_{1,i,t}, w_2 x_{2,i,t})$$

Quantitative Evaluation Engine: Test the Proposal

Purpose

The Quantitative Evaluation Engine **deterministically backtests** each proposed alpha signal against historical forward returns.

Information provided

- ▶ Alpha proposal:

$$\alpha_{i,t} = f(w_1 x_{1,i,t}, w_2 x_{2,i,t})$$

- ▶ Historical discovery data
- ▶ Realized 5-day forward returns

Output: metrics

For each date, stocks are ranked by $\alpha_{i,t}$ and compared with their realized forward returns.

- ▶ **Mean Rank IC**
Did the signal correctly rank future relative returns?
- ▶ **Sharpe ratio**
Did the ranking produce consistent long–short returns?
- ▶ **Mean turnover**
How much did portfolio positions change over time?

Critic Agent: Diagnose the Weakness

Purpose

The Critic Agent interprets the **metrics from the evaluation engine** and identifies the main weakness of the current alpha proposal.

Information provided

- ▶ Current alpha proposal:
 - $\alpha_{i,t} = f(w_1x_{1,i,t}, w_2x_{2,i,t})$
 - Economic thesis
- ▶ Deterministic evaluation metrics:
 - Mean Rank IC
 - Sharpe ratio
 - Mean turnover

Output: critique

The Critic returns:

- ▶ Main weakness
- ▶ Suggests new alpha:
 $\alpha_{i,t} = f(w_1x_{1,i,t}, w_2x_{2,i,t})$
- ▶ Reasoning for the revision

Improver Agent: Revise the Hypothesis

Purpose

The Improver Agent converts the **critique from the Critic** into the next executable alpha proposal in the iterative research loop.

Information provided

- ▶ Current alpha proposal:
 - $\alpha_{i,t}$
 - Economic thesis
- ▶ Evaluation metrics:
 - Mean Rank IC
 - Sharpe ratio
 - Turnover
- ▶ The critique from the Critic Agent

Output: improved proposal

The Improver returns:

- ▶ Final $\alpha_{i,t}$ to use on next iteration
- ▶ Economic thesis

Role in the loop

The improved proposal is returned to the **Quantitative Evaluation Engine**, beginning the next iteration.

Example: Researcher Generates an Initial Hypothesis

Researcher Proposal

Economic thesis:

"Risk-adjusted medium-term momentum: favor assets with strong 20-day momentum relative to their recent volatility, which may indicate persistent trends with better risk efficiency."

Structured proposal

Feature 1: momentum_20d
Feature 2: volatility_20d
Operation: divide
Weight 1: 1.0
Weight 2: 1.0

Executable alpha

$$\alpha_{i,t} = \frac{\text{momentum}_{20d,i,t}}{\text{volatility}_{20d,i,t}}$$

Stocks receive one score per day and are ranked cross-sectionally by this signal.

At this stage

This is a **hypothesis**, not evidence of alpha. The proposal must now be tested by deterministic code.

Example: Evaluation of Initial Hypothesis

Hypothesis being tested

$$\alpha_{i,t} = \frac{\text{momentum}_{20d,i,t}}{\text{volatility}_{20d,i,t}}$$

Discovery-set evaluation

Mean Rank IC	−0.0453
Sharpe	0.697
Mean turnover	0.570

What does the evidence say?

- ▶ Expected: **high risk-adjusted momentum should rank future outperformers higher**
- ▶ Observed: **the cross-sectional relationship is significantly negative**
- ▶ Interpretation: **the proposed momentum direction is not supported by the data**

Example: Critique, Revision, and Re-evaluation

Critic Agent

Diagnosis

"The negative Mean IC and Rank IC indicate that volatility-scaled momentum is directionally misaligned. Division by volatility may also destabilize the ranking."

Suggested response

Preserve momentum as the primary signal, but replace division with a **modest volatility penalty**.

Re-evaluation

	Initial	Revised
Rank IC	-0.0453	→ -0.0524

Revision failed → Performance worsened → **continue the Critic-Improver loop.**

Improver Agent

Revised hypothesis

"Emphasize medium-term momentum while applying a modest volatility penalty, avoiding denominator sensitivity and preserving trend information."

$$\alpha_{i,t}^{\text{new}} = \text{momentum}_{20d,i,t} - 0.25 \text{volatility}_{20d,i,t}$$

Example: Complete Agentic Research Run

Five-iteration discovery budget

The previous slides showed the first transition in detail. Below is the complete bounded research trajectory.

Iter.	Signal / Research Idea	Rank IC	Sharpe	Turnover
1	Momentum divided by volatility	−0.0453	0.697	0.570
2	Momentum minus volatility penalty	−0.0524	−0.862	0.582
3	Inverted momentum + volatility adjustment	0.0524	0.862	0.582
4	Inverted momentum × relative volume	0.0496	0.157	0.627
5	5-day return × relative volume	0.0300	0.300	0.964

Selected discovery winner

$$\alpha_{i,t} = -\text{momentum}_{20d,i,t} + 0.25 \text{ volatility}_{20d,i,t}$$

Rank IC: +0.0524 Sharpe: +0.862 Turnover: 0.582

Validation stage

Rank IC: −0.0400

Sharpe: −1.165

Turnover: 0.454

Decision: **REJECT**

Result The adaptive loop found a stronger signal on discovery data, but the **frozen winner reversed out-of-sample and failed validation.**

Possible Improvements to Implement

Key lesson

Discovery performance improved, but the selected signal **failed out-of-sample validation**.

Scale the research

- ▶ **Scale the number of iterations allowed**
 - More observations during discovery stage
 - Allow more stocks
- ▶ **Produce more interesting alpha signals**
 - Use more independent raw data to construct features
 - Extend to a richer set of operations
- ▶ **Split discovery data into folds**
 - Expose performance differences across time periods
- ▶ **LLM-derived alternative data**
 - Convert financial text into a feature that can be used to construct an alpha signal

Improve robustness

- ▶ **Walk-forward testing**
 - Test across multiple time periods
- ▶ **Stability requirements**
 - Require consistent performance across market regimes
- ▶ **Transaction costs**
 - Evaluate net, not just gross, performance

Conclusions

Summary of Prototype

LLM agents **propose, critique, and revise** alpha hypotheses; deterministic code **evaluates** them.

Key takeaways

1. **Agents can learn from empirical feedback.**
Failed hypotheses inform subsequent proposals.
2. **Discovery performance is not enough.**
Our strongest discovery signal failed protected validation.
3. **Robust evaluation is essential.**
Deterministic testing and protected data constrain overfitting.